Fangzhou "Bonnie" Dong

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Chandler, AZ

Education

Arizona State University Tempe, AZ

Ph.D. Computer Science Jan 2020—Present

- Research topics: system security, program abstraction, symbolic execution, reverse engineering
- Supervisor: Dr. Fish Wang

Sichuan University Chengdu, China

B.E. Information Security Sep 2015—June 2019

- Ranking: 1/74, GPA: 3.77/4.0
- Thesis: Research on Program Data Structure Inference Techniques Based on Binary Code

Publications

Discovering Blind-Trust Vulnerabilities in PLC Binaries via State Machine Recovery

Fangzhou Dong, Arvind S Raj, Efrén López-Morales, Siyu Liu, Yan Shoshitaishvili, Tiffany Bao, Adam Doupé, Muslum Ozgur Ozmen and Ruoyu Wang

Proceedings of Network and Distributed System Security (NDSS) Symposium 2026.

Fuzz to the Future: Uncovering Occluded Future Vulnerabilities via Robust Fuzzing

Arvind S Raj, Wil Gibbs, Fangzhou Dong, Jayakrishna Menon Vadayath, Michael Tompkins, Steven Wirsz, Yibo Liu, Zhenghao Hu, Chang Zhu, Gokulkrishna Praveen Menon, Brendan Dolan-Gavitt, Adam Doupé, Ruoyu Wang, Yan Shoshitaishvili, and Tiffany Bao

Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS '24).

Operation Mango: Scalable Discovery of Taint-Style Vulnerabilities in Binary Firmware Services

Wil Gibbs, Arvind S Raj, Jayakrishna Menon Vadayath, Hui Jun Tay, Justin Miller, Akshay Ajayan, Zion Leonahenahe Basque, Audrey Dutcher, and Fangzhou Dong, Xavier Maso, Giovanni Vigna, Christopher Kruegel, Adam Doupé, Yan Shoshitaishvili, and Ruoyu Wang

Proceedings of the 33nd USENIX Conference on Security Symposium (USENIX Sec '24).

Greenhouse: Single-Service Rehosting of Linux-Based Firmware Binaries in User-Space Emulation

Hui Jun Tay, Kyle Zeng, Jayakrishna Menon Vadayath, Arvind S Raj, Audrey Dutcher, Tejesh Reddy, Wil Gibbs, Zion Leonahenahe Basque, Fangzhou Dong, Zack Smith, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, and Ruoyu Wang

Proceedings of the 32nd USENIX Conference on Security Symposium (USENIX Sec '23).

Experience

SEFCOM Lab, Arizona State University

Graduate Research Assistant Jan 2020—Present

- Led research project of an automated framework to discover logic bugs in ICS controller binaries via state machine recovery.

- Reverse engineer and debug software/firmware using IDA Pro, Ghidra, GDB and angr.
- Contribute to research projects spanning vulnerability discovery, binary program analysis and fuzzing.
- Actively contribute to open-source binary analysis project angr.

Core Contributor—DARPA FIRE Program Feb 2024—Jan 2026

- Reverse engineered CPS firmware (drones and rovers) to recover systems' behaviors, components, interfaces and key algorithms.
- Built and delivered behavior models from reverse-engineered firmware to enable system understanding and support vulnerability and backdoor assessment.

Mentoring & Teaching

Mentor, High School Summer Internship Program May—July 2023, 2024

Teaching Assistant, CSE 545 Software Security 2021 Fall, 2023 Spring

Activities

Attendee. Cyber Truck Challenge 2023, 2024

Member, DEFCON CTF quals & finals with team Shellphish 2022, 2023, 2025

Member, Pwn2Own Toronto 2022 with team SEFCOM T0 2022